step 1: Configuration
1. git --version
2. git config --global user.name "Leela1325"
    used for configuring name
    opposite: git config --global --unset user.name
3. git config --global user.email "leelaprasad@gmail.com"
    used for cinfiguring the email of github
    opposite: git config --global --unset user.email
4. git config --global init.defaultBranch main
    used for creating a default branch. Git needs a starting branch to start the version control
    opp: git config --global --unset init.defaultBranch
5. git config --global --list
    used for checking the configurations
6. git help

step 2: Working

1.git init
    used for initalising/reinitialising the git for a folder
    opp: rm -rf .git
2. touch filename.filetypeextension
    used for creating a file
    opp: rm filename.filetypeextension
3. echo "Data" > filename.filetypeextension
    used for add date in the file
4. git status
5. git add filename
    used for put the file to storage area from working
    opp: git reset filename
6. git add .
    used for add all files
    oppL git reset
7. git commit -m "content"
    used for commit the changes
    opp: two cases
    if root is current commit : git update-ref -d HEAD (deletes the current root/Top root)
    else if : git reset --soft HEAD~1 (goes 1 commit back and keep all the files and changes as it is)
    else if: git reset --mixed HEAD~1(goes1 step back and removes files from storing area and keep changes as it is)
    else : git reset --hard HEAD~1 (remove all changes and deletes the files that we created in latest commit)
8. git log

used for view commit history

9. git diff

used for all the changes that we did

10. git diff --staged

used for view the changes of storage area (such as added latest files)

11. git diff HEAD~1 HEAD

used for check the diffference of previour commit and latest commit.

12. git checkout -- filename.ext

used for delete the changes of the particular file..like undo of a content of files..

Step 3: Branch Commands

1.git branch

used for list the branches present in the folder

* symbol tells us the current branch

2. git branch branch_name

used for create a new branch

A branch is a paraller version of a main/another verison of codes

opp: git branch -d branch_name (deletes the branch)

3. ls

lists all the files of branch

4. git checkout branch_name

used for switch to respect branch

5. git merge branch_name

merges a branch to current branch..(This command has to be use at current branch ..not in merging branch)

opp: git reset --hard HEAD~1

step 4: Remote

1.git remote add origin "github repository link"

used for connecting to repository

origin is nickname for our repository

remote means github repository

opp: git remote remove origin

2. git remote -v

used for verification of remote connection

3. git push -u origin main

used for push the latest commit to github

-u is used for act as remember me..when we use first time, we are telling git that we are push/pull only from this repository

4. git pull origin main

downloads the files from github and automatically merges in our local files.

5. git push origin branch_name

used for push the latest commits..used when we are not pushing first time so that we can avoid -u

6. rm filename

used for deleteing the file in locol depository.

How to delete a file in github..

A) we cannot delete in github. So we use rm filename to delete in localfolder and we do git add . and git commit so that it replace all changes from local to github

7. git push origin --delete branch_name

used for deleting a branch remote (deletes in github)

Stash: just a stack

git stash - to save work temporarly

git stash push -m "" - to save with description

git stash -u -includes untracked files

git stash list - displays all the stash of stack

git stash pop -removes and gives the latest stash

git stash apply- only applies the latest stash

git stash drop - deletes latest stash

git stash clear -clears all the stack